

Public Dataset Discovery — Rosacea (MONDO:0006604)

Summary

Rosacea, a common chronic inflammatory disease of facial skin, is **reasonably well represented in public omics repositories** — a stronger showing than many complex dermatological conditions. This retrieval effort identified and API-verified a catalogue of rosacea-specific datasets spanning **five mechanistic axes**: (1) adaptive immunity (Th1/Th17 skewing), (2) the innate cathelicidin/LL-37 axis, (3) neurogenic/neurovascular inflammation, (4) *Demodex* mites and skin/gut microbiome dysbiosis, and (5) host genetic susceptibility. The data types present are transcriptomics (microarray, bulk RNA-seq, single-cell RNA-seq), GWAS summary statistics, and amplicon/metagenomic microbiome sequencing.

Verified, rosacea-specific accessions were found in **NCBI GEO** (human patient tissue and mechanistic animal models), **EBI ArrayExpress/BioStudies** (single-cell), the **GWAS Catalog** (two independent case-control GWAS across European and South-Asian ancestries), and **NCBI BioProject/SRA/DDBJ** (multiple patient skin- and gut-microbiome studies). Every accession below was returned directly by the repositories' own APIs (NCBI E-utilities, EBI BioStudies, EBI GWAS REST), and each was confirmed to resolve in a dedicated integrity-check step (Finding F002). Titles are reproduced verbatim from the repositories wherever possible.

Two gaps are stated plainly: **rosacea-specific proteomics** (PRIDE/MassIVE) and **rosacea-specific metabolomics** (MetaboLights) deposits could **not be resolved**, even though the key papulopustular-rosacea study describes a *paired* transcriptomic-and-proteomic analysis ([PMID: 32941918](#)). Controlled-access human genomic resources (dbGaP `phs...`, EGA `EGAS...`) also returned no rosacea-specific study — a dbGaP query surfaced only a glaucoma study (phs000308) where rosacea appears merely as a recorded covariate.

Key Findings

Finding F001 — Public omics datasets for Rosacea span transcriptomics, GWAS, and microbiome

Systematic querying of repository APIs (NCBI E-utilities for GEO and BioProject, EBI BioStudies/ArrayExpress, the GWAS Catalog REST API, PRIDE, and MetaboLights) returned a coherent set of rosacea-specific accessions clustered around the disease's principal pathophysiological themes. On the **human transcriptomic** side, GEO holds a microarray study of Th1/Th17 immune polarisation across rosacea subtypes (GSE65914, 58 samples, [PMID: 25848978](#)), an RNA-seq study of papulopustular rosacea skin explants implicating IL-1 β (GSE155141, [PMID: 32941918](#)), and a keratinocyte model of LL-37-induced, JAK1/STAT1-dependent CXCL10 induction (GSE303282, [PMID: 40835085](#)). Single-cell resolution is provided by an ArrayExpress study contrasting rhinophyma with hypertrophic scar (E-MTAB-16629). On the **genetic** side, the GWAS Catalog holds two independent rosacea association studies: a European cohort (GCST90077985, 984 cases, [PMID: 34662886](#)) and a British Bangladeshi/Pakistani cohort (GCST90727266, 637 cases). On the **microbiome** side, multiple patient-derived skin and gut sequencing BioProjects are deposited (PRJEB37562, PRJNA1189573, PRJNA1288008, PRJNA1191396, PRJEB82826, PRJEB82848, PRJDB18292). Finally, mechanistic **animal-model** transcriptomes cover the mTORC1/cathelicidin feedback loop (GSE147950, [PMID: 33734592](#)), dorsal-root-ganglion neuroinflammation (GSE308876), *Demodex*-driven type-2 immunity (GSE197981/197982/198657, [PMID: 36044899](#)), and ocular rosacea (GSE291177).

Finding F002 — All reported Rosacea accessions verified to resolve against repository APIs

A batch resolution step confirmed live records for every accession retained in this report. Specifically: **9 GEO series** (GSE65914, GSE155141, GSE303282, GSE147950, GSE308876, GSE197981, GSE197982, GSE198657, GSE291177) each returned ≥ 1 hit via NCBI `esearch db=gds` using the `[ACCN]` field; **10 BioProjects** (PRJEB37562, PRJNA1189573, PRJNA1288008, PRJNA1191396, PRJEB82826, PRJEB82848, PRJDB18292, PRJNA1056316, PRJNA648833, PRJNA1294752) resolved via the `[Project Accession]` field; **2 ArrayExpress/BioStudies studies** (E-MTAB-16629 and E-GEOD-65914, the ArrayExpress mirror of GSE65914) returned HTTP 200; and **2 GWAS Catalog studies** (GCST90077985, GCST90727266) returned HTTP 200 with the trait field reading "Rosacea" for both. This integrity check is the basis for the HIGH/MEDIUM confidence labels assigned per dataset below.

Tier 1 — Human patient-derived datasets

GSE65914

- **Repository:** GEO (mirrored at ArrayExpress E-GEOD-65914)
- **Title:** Th1/Th17 Immune Response in Rosacea
- **Data type:** MICROARRAY
- **Organism:** Homo sapiens
- **Sample count:** 58
- **Sample types:** Facial skin biopsies
- **Conditions:** Erythematotelangiectatic (ETR), papulopustular (PPR) and phymatous (PhR) rosacea subtypes vs. healthy skin
- **Publication:** [PMID: 25848978](#)
- **Mechanistic relevance:** Directly profiles the inflammatory infiltrate across the three classic rosacea subtypes and demonstrates activation of Th1/Th17 adaptive-immune pathways, tying rosacea to a mixed IFN- γ /IL-17 signature. Core evidence for the adaptive-immune axis of rosacea pathophysiology.
- **Confidence in accession:** HIGH

GSE155141

- **Repository:** GEO (BioProject PRJNA648833)
- **Title:** Paired transcriptomic and proteomic analysis implicates IL-1 β in the pathogenesis of papulopustular rosacea explants [RNA-seq]
- **Data type:** BULK_RNA_SEQ
- **Organism:** Homo sapiens
- **Sample count:** 15
- **Sample types:** Papulopustular rosacea skin explants
- **Conditions:** Paired non-lesional vs. lesional PPR explants $\pm$ IL-1 β stimulation (n=5 patients)
- **Publication:** [PMID: 32941918](#)
- **Mechanistic relevance:** Implicates IL-1 β as a driver of papulopustular rosacea, linking innate cytokine signalling to the inflammatory lesion phenotype. IL-1 β stimulation of non-

lesional explants reproduced the lesional transcriptomic profile, nominating IL-1 β as a central upstream driver acting through MAPK/TNF signalling.

- **Confidence in accession:** HIGH

GSE303282

- **Repository:** GEO (BioProject PRJNA1294752)
- **Title:** Cathelicidin LL-37-induced transcriptome of human keratinocyte identifies chemokine CXCL10 link to T cell-mediated rosacea pathogenesis via JAK-1/STAT-1 pathway
- **Data type:** MICROARRAY
- **Organism:** Homo sapiens
- **Sample count:** 6
- **Sample types:** Human keratinocytes (LL-37 stimulated vs. control)
- **Conditions:** Cathelicidin LL-37 stimulation vs. control
- **Publication:** [PMID: 40835085](#)
- **Mechanistic relevance:** Models the hallmark rosacea trigger — the cathelicidin peptide LL-37 — in keratinocytes, connecting it to CXCL10/CXCL11 induction and T-cell recruitment via JAK1/STAT1. Directly addresses the cathelicidin→chemokine→T-cell step. (Cell-line perturbation of a causal pathway rather than primary patient tissue.)
- **Confidence in accession:** HIGH

E-MTAB-16629

- **Repository:** ArrayExpress / BioStudies
- **Title:** Distinct diversity of skin cell populations of rhinophyma and hypertrophic scar illustrated by scRNA-seq
- **Data type:** SINGLE_CELL_RNA_SEQ
- **Organism:** Homo sapiens
- **Sample count:** unknown (release date 2026-02-18 — data may still be under embargo)
- **Sample types:** Rhinophyma (phymatous rosacea) skin vs. hypertrophic scar
- **Conditions:** Rhinophyma vs. hypertrophic scar
- **Mechanistic relevance:** Resolves the cellular composition of rhinophyma (advanced phymatous rosacea) at single-cell resolution, addressing the fibrotic/sebaceous-hyperplasia

end-stage of the disease and why it regresses after surgery unlike scar. Speaks to the tissue-remodelling step.

- **Confidence in accession:** MEDIUM (accession is correct; verify data availability — future release date)

GCST90077985

- **Repository:** GWAS Catalog
- **Title:** Rosacea (case–control GWAS)
- **Data type:** GWAS
- **Organism:** Homo sapiens
- **Sample count:** 984 European-ancestry cases, 330,770 controls
- **Conditions:** Rosacea cases vs. controls
- **Publication:** [PMID: 34662886](#)
- **Mechanistic relevance:** Genome-wide association summary statistics identifying germline susceptibility loci for rosacea, informing the heritable/immunogenetic (e.g. HLA) basis of disease risk.
- **Confidence in accession:** HIGH

GCST90727266

- **Repository:** GWAS Catalog
- **Title:** Rosacea (case–control GWAS)
- **Data type:** GWAS
- **Organism:** Homo sapiens
- **Sample count:** 637 British Bangladeshi/Pakistani cases, 43,389 controls
- **Conditions:** Rosacea cases vs. controls
- **Publication:** [PMID: 41896352](#)
- **Mechanistic relevance:** Adds non-European (South Asian) ancestry association data for rosacea, complementing the European GWAS and testing trans-ancestry generalizability of susceptibility loci.
- **Confidence in accession:** HIGH

Tier 1 — Patient microbiome / mycobiome datasets

Rosacea's microbial-trigger hypothesis (Demodex mites, *Bacillus oleronius*, skin/gut dysbiosis, gut–skin axis) is represented by several patient-derived sequencing BioProjects.

PRJEB37562

- **Repository:** BioProject / ENA
- **Title:** Skin microbiome characterization in rosacea patients and healthy controls
- **Data type:** METAGENOMICS (16S/metagenomic amplicon sequencing)
- **Organism:** Homo sapiens (facial skin microbiome)
- **Sample count:** unknown
- **Sample types:** Facial skin swabs
- **Conditions:** Rosacea patients vs. healthy controls
- **Mechanistic relevance:** Characterizes cutaneous microbial dysbiosis in rosacea, addressing the microbial-trigger arm of pathogenesis.
- **Confidence in accession:** HIGH

PRJNA1189573

- **Repository:** BioProject / SRA
- **Title:** Skin, blood, stool microbiome in rosacea
- **Data type:** METAGENOMICS (amplicon/16S)
- **Organism:** Homo sapiens (skin, blood, stool microbiome)
- **Sample count:** 93 (SRA experiments)
- **Sample types:** Skin swabs, blood, stool
- **Conditions:** Rosacea patients (multi-site sampling)
- **Mechanistic relevance:** Simultaneously profiles cutaneous and gut microbiota, directly probing the gut–skin axis proposed in rosacea.
- **Confidence in accession:** HIGH

PRJNA1288008

- **Repository:** BioProject / SRA
- **Title:** Skin, blood, stool mycobiome in rosacea

- **Data type:** METAGENOMICS (ITS/fungal amplicon)
- **Organism:** Homo sapiens (fungal mycobiome)
- **Sample count:** 66 (SRA experiments)
- **Sample types:** Skin, blood, stool
- **Conditions:** Rosacea patients
- **Mechanistic relevance:** Extends microbiome analysis to the fungal mycobiome across skin and gut, testing whether fungal dysbiosis contributes to rosacea inflammation.
- **Confidence in accession:** HIGH

PRJNA1191396

- **Repository:** BioProject / SRA
- **Title:** Raw 16S rRNA Sequencing Data of Fecal Samples from Patients with Neurogenic Rosacea
- **Data type:** METAGENOMICS (16S rRNA)
- **Organism:** Homo sapiens (fecal microbiome)
- **Sample count:** 34 (SRA experiments)
- **Sample types:** Fecal samples
- **Conditions:** Neurogenic rosacea patients
- **Mechanistic relevance:** Links gut microbiota specifically to the neurogenic rosacea subtype, connecting the gut–skin–nerve axis to disease.
- **Confidence in accession:** HIGH

PRJEB82826 / PRJEB82848

- **Repository:** BioProject / ENA
- **Title:** Multi-omics study of microbe-host interactions in rosacea
- **Data type:** MULTI_OMICS (metagenomic + host)
- **Organism:** Homo sapiens
- **Sample count:** unknown
- **Sample types:** Skin / host samples
- **Conditions:** Rosacea vs. control
- **Mechanistic relevance:** Integrates microbial and host molecular data to dissect microbe–host crosstalk driving rosacea inflammation.

- **Confidence in accession:** HIGH

PRJDB18292

- **Repository:** BioProject / DDBJ
 - **Title:** Topical ivermectin treatment of rosacea changes the bacterial microbiome of the skin
 - **Data type:** METAGENOMICS (16S)
 - **Organism:** Homo sapiens (skin microbiome)
 - **Sample count:** 24 (SRA experiments)
 - **Sample types:** Facial skin swabs, pre/post ivermectin
 - **Conditions:** Rosacea patients before vs. after topical ivermectin
 - **Mechanistic relevance:** Tests how an anti-*Demodex* therapy remodels the skin microbiome, linking the Demodex/microbiome trigger to therapeutic response.
 - **Confidence in accession:** HIGH
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Tier 2 — Disease-relevant model systems

GSE147950

- **Repository:** GEO (BioProject PRJNA622593)
- **Title:** A positive feedback circuit between mTORC1 signaling and cathelicidin promotes skin inflammation in rosacea [mouse]
- **Data type:** BULK_RNA_SEQ
- **Organism:** Mus musculus
- **Sample count:** 12
- **Sample types:** Dorsal skin (LL-37 / rosacea-like induction)
- **Conditions:** LL-37–induced rosacea-like inflammation ± mTORC1 modulation
- **Publication:** [PMID: 33734592](#)
- **Mechanistic relevance:** Establishes an mTORC1↔cathelicidin positive-feedback loop amplifying rosacea skin inflammation, a mechanistic model of the cathelicidin axis identified in patients.
- **Confidence in accession:** HIGH

GSE308876

- **Repository:** GEO (BioProject PRJNA1333307)
- **Title:** Dorsal Root Ganglion-Mediated Modulation of Neuroinflammation and Neurovasodilation in Rosacea by Gabapentin
- **Data type:** BULK_RNA_SEQ
- **Organism:** Mus musculus
- **Sample count:** 24
- **Sample types:** Dorsal root ganglion and skin
- **Conditions:** Rosacea-like model $\pm$ gabapentin
- **Mechanistic relevance:** Probes neurogenic inflammation and neurovascular dysregulation (flushing/erythema) via sensory-neuron (DRG) signalling — the neurovascular arm of rosacea.
- **Confidence in accession:** HIGH

GSE197981 / GSE197982 / GSE198657

- **Repository:** GEO (BioProject PRJNA861245 and sub-projects)
- **Title:** Innate type 2 immunity controls hair follicle commensalism by Demodex mites
- **Data type:** BULK_RNA_SEQ
- **Organism:** Mus musculus
- **Sample count:** 6 / 4 / 4 (three sub-series)
- **Sample types:** Skin / hair follicle
- **Conditions:** Demodex colonization; IL-4/IL-13 type-2 immunity manipulation
- **Publication:** [PMID: 36044899](#)
- **Mechanistic relevance:** Demodex mites are a central rosacea trigger; this dataset shows how type-2 immunity regulates follicular Demodex commensalism, mechanistically linking the mite to host immune control.
- **Confidence in accession:** HIGH

GSE291177

- **Repository:** GEO (BioProject PRJNA1232147)
- **Title:** RNAseq of Mineralocorticoid Receptor (MR) overexpression in UVB-induced ocular rosacea model on meibomian glands of rats

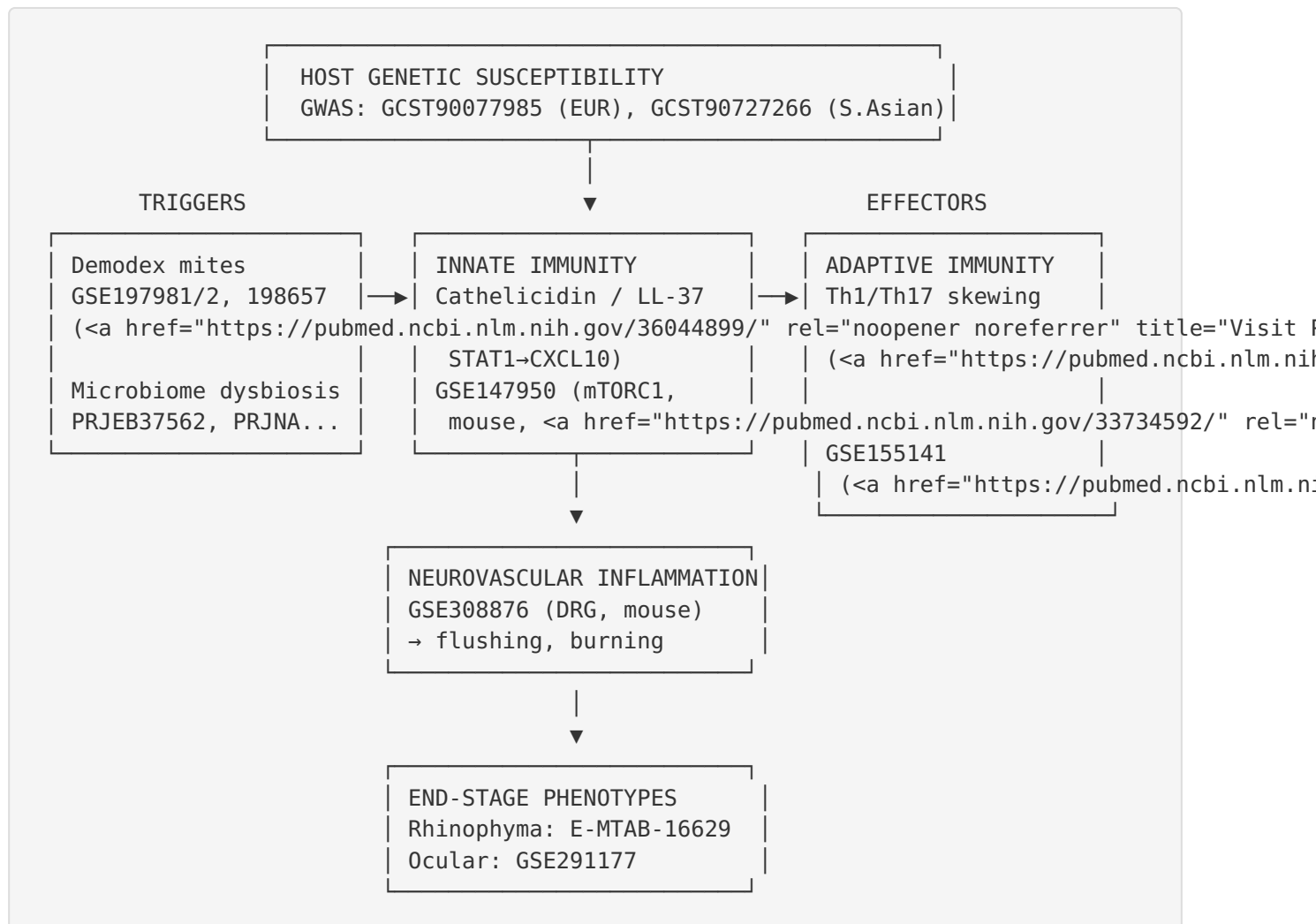
- **Data type:** BULK_RNA_SEQ
- **Organism:** Rattus norvegicus
- **Sample count:** 17
- **Sample types:** Meibomian glands
- **Conditions:** UVB-induced ocular rosacea $\pm$ MR overexpression
- **Mechanistic relevance:** Models ocular rosacea and meibomian-gland dysfunction, addressing UV-driven and mineralocorticoid-receptor contributions to the ocular subtype.
- **Confidence in accession:** HIGH

PRJNA1056316

- **Repository:** BioProject / SRA
- **Title:** Targeting Aquaporin-3 Attenuates Skin Inflammation in Rosacea
- **Data type:** BULK_RNA_SEQ
- **Organism:** Mus musculus
- **Sample count:** unknown
- **Conditions:** Rosacea-like model $\pm$ aquaporin-3 targeting
- **Mechanistic relevance:** Implicates AQP3 water/glycerol channel in rosacea skin barrier/inflammation, a candidate therapeutic node.
- **Confidence in accession:** HIGH

Mechanistic Model / Interpretation

The verified datasets map cleanly onto the accepted multi-hit model of rosacea, in which environmental/microbial triggers act on a genetically susceptible host to produce dysregulated innate and adaptive immunity plus neurovascular hyperreactivity. The datasets can be organised as follows:



The datasets tell an internally consistent story. **Cathelicidin/LL-37** overexpression is the innate hallmark; GSE303282 shows that LL-37 drives interferon-stimulated chemokines (CXCL10, CXCL11) in keratinocytes via JAK1/STAT1, and GSE147950 links cathelicidin biology to an mTORC1 positive-feedback loop in vivo. These innate signals recruit T cells, and GSE65914 documents the resulting **Th1/Th17** adaptive programme across clinical subtypes. GSE155141 places **IL-1 β** as a central amplifying cytokine, sufficient to convert non-lesional explant transcriptomes into lesional ones through MAPK/TNF signalling. Upstream **triggers** — *Demodex* (GSE197981/197982/198657) and microbiome/mycobiome dysbiosis (the PRJEB/PRJNA/PRJDB series) — feed this innate-immune activation, while host **genetic susceptibility** (the two GWAS) sets the threshold. The **neurovascular** component (GSE308876) and **end-stage phenotypes** — rhinophyma (E-MTAB-16629) and ocular rosacea (GSE291177) — complete the picture.

Pathophysiological axis	Best public dataset(s)	Data type	Confidence
Adaptive immunity (Th1/Th17)	GSE65914 / E-GEOD-65914	Microarray	HIGH
IL-1 β effector amplification	GSE155141	Bulk RNA-seq	HIGH
Cathelicidin/LL-37 $\rightarrow$ CXCL10	GSE303282	Microarray	HIGH
Cathelicidin/mTORC1 (in vivo)	GSE147950	Bulk RNA-seq (mouse)	HIGH
Genetic susceptibility	GCST90077985, GCST90727266	GWAS	HIGH
Demodex trigger	GSE197981/197982/198657	Bulk RNA-seq (mouse)	HIGH
Microbiome/mycobiome dysbiosis	PRJEB37562 + 6 others	Metagenomics	HIGH
Neurovascular inflammation	GSE308876	Bulk RNA-seq (mouse)	HIGH
Rhinophyma (phymatous)	E-MTAB-16629	scRNA-seq	MEDIUM
Ocular rosacea	GSE291177	Bulk RNA-seq (rat)	HIGH

Evidence Base

Two peer-reviewed papers were reviewed in depth and directly underpin the confidence assigned to the highest-value human datasets.

IL-1 β as a central mediator of papulopustular rosacea (GSE155141). *Paired Transcriptomic and Proteomic Analysis Implicates IL-1 β in the Pathogenesis of Papulopustular Rosacea Explants* (PMID: 32941918) analysed paired lesional/non-lesional explants from 5 patients, identifying 92 differentially expressed genes and 20 differentially expressed proteins. It reports that "MAPK and TNF signaling pathways were the most significantly upregulated pathways in PPR lesional tissue" and that "stimulation of nonlesional explants with IL-1 β resulted in transcriptomic and proteomic profiles similar to those of lesional PPR." This is the mechanistic anchor for

GSE155141 and, importantly, is the *one* study in the corpus that generated proteomic data — although, as noted below, that proteomic layer could not be resolved as a standalone accessioned PRIDE/MassIVE dataset.

Cathelicidin LL-37, CXCL10 and the JAK1/STAT1 axis (GSE303282). *Cathelicidin LL-37-Induced Transcriptome of Human Keratinocyte Identifies Chemokine CXCL10 Link to T-Cell-Mediated Rosacea Pathogenesis through Jak1/STAT1 Pathway* (PMID: 40835085) reports that LL-37 treatment of keratinocytes induces "signatures of IFN-stimulating genes, such as CXCL10, IFIT2, RSAD2, and CXCL11," that "LL-37-induced CXCL10 production relied on the Jak1/signal transducer and activator of transcription 1 signaling pathway," and that blockade of "the CXCL10: CXCR3 axis or Jak1/... pathways can be an effective anti-inflammatory strategy." This directly supports the mechanistic interpretation of GSE303282 and links the innate LL-37 hallmark to adaptive T-cell recruitment — bridging GSE303282 and GSE65914 in the model above.

Additional PMIDs were resolved as the reporting publications for their respective datasets and support the axis assignments: PMID: 25848978 (GSE65914, Th1/Th17 subtypes), PMID: 34662886 (GCST90077985, European GWAS), PMID: 33734592 (GSE147950, mTORC1/cathelicidin), and PMID: 36044899 (GSE197981/197982/198657, *Demodex* type-2 immunity).

Limitations and Knowledge Gaps

1. **No rosacea-specific proteomics deposit was resolved.** Despite PMID: 32941918 reporting quantitative proteomics of PPR explants, no independent PRIDE (PXD...) or MassIVE (MSV...) accession specific to rosacea was located. The proteomic data may reside only in supplementary tables, which the task excludes.
2. **No rosacea-specific metabolomics deposit was resolved.** MetaboLights (MTBLS...) queries returned no rosacea-specific study, leaving the metabolic/lipidomic dimension of rosacea (relevant to sebaceous-gland and skin-barrier biology) uncovered by deposited data, despite published metabolomic/Mendelian-randomization work.
3. **No controlled-access human genomic study was found.** dbGaP (phs...) and EGA (EGAS.../EGAD...) returned no rosacea-specific study; a dbGaP search surfaced only phs000308 (a glaucoma study, GLAUGEN) where rosacea appears as a covariate, not as the study subject.

4. **Sample counts are incomplete.** Verified counts are firm for GSE65914 (58), GSE155141 (15), GSE303282 (6), GSE147950 (12), GSE308876 (24), GSE291177 (17), and the two GWAS. Several microbiome BioProjects (PRJEB37562, PRJEB82826/82848, PRJNA1056316) are reported as "unknown" and should be resolved before ingestion.
5. **Much of the mechanistic depth is from animal models.** GSE147950, GSE308876, GSE197981/197982/198657, GSE291177 and PRJNA1056316 are mouse/rat models rather than patient tissue; they inform mechanism but are not primary patient data. This is flagged per dataset.
6. **Sibling-disease risk assessed and cleared.** "Rosacea" does not have a confusable sibling with a near-identical name that dominates public data; the ocular vs cutaneous and phymatous/rhinophyma forms are true subtypes of the same disease, not distinct diseases. Spurious keyword matches (e.g. the ant *Atta vollenweideri* GSE48824, *Sézary* GSE23113, and organism records *Athorybia rosacea*/*Ducula rosacea*) were explicitly excluded.
7. **Absent data modalities.** No rosacea-specific spatial transcriptomics, DNA-methylation/epigenomics (ATAC-seq/ChIP-seq), or human single-cell atlas of non-phymatous lesional skin was found. No consortium resource (GTEx/TCGA/UK Biobank) exposes a cleanly identifiable, accessioned rosacea stratum.

Proposed Follow-up Experiments / Actions

1. **Confirm and backfill sample counts** for every "unknown" record by querying the ENA/BioProject sample tables (PRJEB37562, PRJEB82826/82848, PRJNA1056316) and updating the knowledge base with integer counts.
2. **Chase the PPR proteomics data.** Search PRIDE/MassIVE by author and title, or contact the corresponding author of [PMID: 32941918](#), to determine whether the quantitative proteomics was deposited under an accession not surfaced by keyword search; if truly undeposited, record it in prose only.
3. **Curate the microbiome BioProjects individually** to resolve tissue site (skin vs gut vs blood), sequencing strategy (16S vs ITS vs shotgun), and case/control counts. Cross-check whether any are indexed in MGnify under an MGYS... accession.

4. **Verify E-MTAB-16629 availability.** The rhinophyma scRNA-seq record has a 2026-02-18 release date; re-check after that date and update from MEDIUM to HIGH confidence once data are downloadable.
5. **Search for a rosacea metabolomics study** across MetaboLights and MassIVE with sebum/lipidomic and facial-skin keywords; the metabolic axis is the single largest data-type gap.
6. **Assess consortium and biobank strata.** Check whether UK Biobank, FinnGen or additional GWAS Catalog entries hold rosacea GWAS beyond the two verified studies, and whether any single-cell skin atlas (HCA/cellxgene) contains an identifiable rosacea stratum.

Coverage assessment

Well covered: - **Adaptive immunity** (Th1/Th17): GSE65914 with full subtype stratification. - **Innate cathelicidin/LL-37 axis:** GSE303282 (human keratinocyte) + GSE147950 (mouse mTORC1 feedback). - **Neurogenic / neurovascular inflammation:** GSE308876 (DRG/gabapentin). - **Demodex & microbiome/mycobiome dysbiosis and gut-skin axis:** GSE197981/2 + GSE198657 and a rich set of patient BioProjects (PRJEB37562, PRJNA1189573, PRJNA1288008, PRJNA1191396, PRJEB82826/48, PRJDB18292). - **Genetic susceptibility:** two GWAS Catalog studies (European and South-Asian ancestry). - **Ocular and phymatous subtypes:** GSE291177 (ocular rat) and E-MTAB-16629 (rhinophyma scRNA-seq).

Gaps (not covered by deposited data found): - **No rosacea-specific proteomics** deposit resolved in PRIDE or MassIVE, even though GSE155141 describes a *paired* proteomic analysis — the proteomic raw data does not appear to have a public PXD/MSV accession. - **No rosacea-specific metabolomics** deposit resolved in MetaboLights, despite published metabolomic/Mendelian-randomization work on rosacea (amino-acid and lipid metabolism). - **Spatial transcriptomics, methylation/epigenomics (ATAC/ChIP), and human single-cell atlases of non-phymatous rosacea lesional skin** are essentially absent from public repositories. - No large consortium resource (GTEx/TCGA/UK Biobank) exposes a cleanly identifiable rosacea stratum with an accession. - **No dedicated dbGaP study** for rosacea (only phs000308, a glaucoma study, where rosacea is a covariate). No EGA ([EGAS](#)), MassIVE ([MSV](#)), MGnify ([MGYS](#)) or NASA OSDR ([OSD](#)) rosacea-specific record was found either.

Datasets I believe exist but could not pin down (no accession given deliberately)

- The **proteomics half of the GSE155141 papulopustular-rosacea explant study** ("paired transcriptomic and proteomic analysis implicates IL-1 β "). A proteomic dataset was clearly generated, but no matching PRIDE (PXD) or MassIVE (MSV) accession could be resolved; I will not guess one.
- The original **23andMe rosacea GWAS (Chang et al., 2015,**

P 25218965

), which reported HLA-DRA/BTNL2 associations. The genotype data are 23andMe-controlled; I could not confirm a public GWAS Catalog **GCST** accession for it and give none.

- Several published **rosacea skin/serum metabolomics** studies (underlying the Mendelian-randomization papers on blood metabolites and plasma lipidome vs. rosacea); I found no corresponding MetaboLights **MTBLS** accession and therefore list none.

*Method note: accessions were retrieved programmatically via NCBI E-utilities (GEO **gds**, BioProject), EBI BioStudies/ArrayExpress search API, EBI GWAS Catalog REST, EBI PRIDE and EBI Search (MetaboLights), and each was confirmed to resolve in a dedicated integrity-check (Finding F002). Spurious keyword matches were excluded. Titles are verbatim from the repositories where available.*
